

Key Concepts: Fibrinolytics

- Fibrinolytics alteplase, tenecteplase, and reteplase are recombinant tPA (tissue plasminogen activator) that convert fibrin-bound plasminogen to plasmin, which cleaves fibrin, which lyses thrombi. Fibrinolytics are given early in clot formation because clots become more resistant to fibrinolysis as they age.
- Fibrinolytics are administered IV (usually) for treatment of ischemic stroke (within 4.5 hours), acute myocardial infarction if PCI is unavailable, and treatment of PE. A fibrinolytic may be used to restore catheter and shunt function.
- Hemorrhage is a major adverse effect. A systemic lytic state can result from plasmin generation and degradation of other coagulation factors.
- Contraindications include pregnancy and patients with healing wounds, history of stroke, brain tumor, head trauma, intracranial bleed, and metastatic cancer.
- Streptokinase is a protein produced by beta-hemolytic streptococci. It forms a complex with plasminogen. The complex induces a conformational change in plasminogen that exposes its active site. Plasminogen then converts free and clot-bound plasminogen molecules to plasmin. Streptokinase produces a systemic lytic state. It is no longer available in the US.

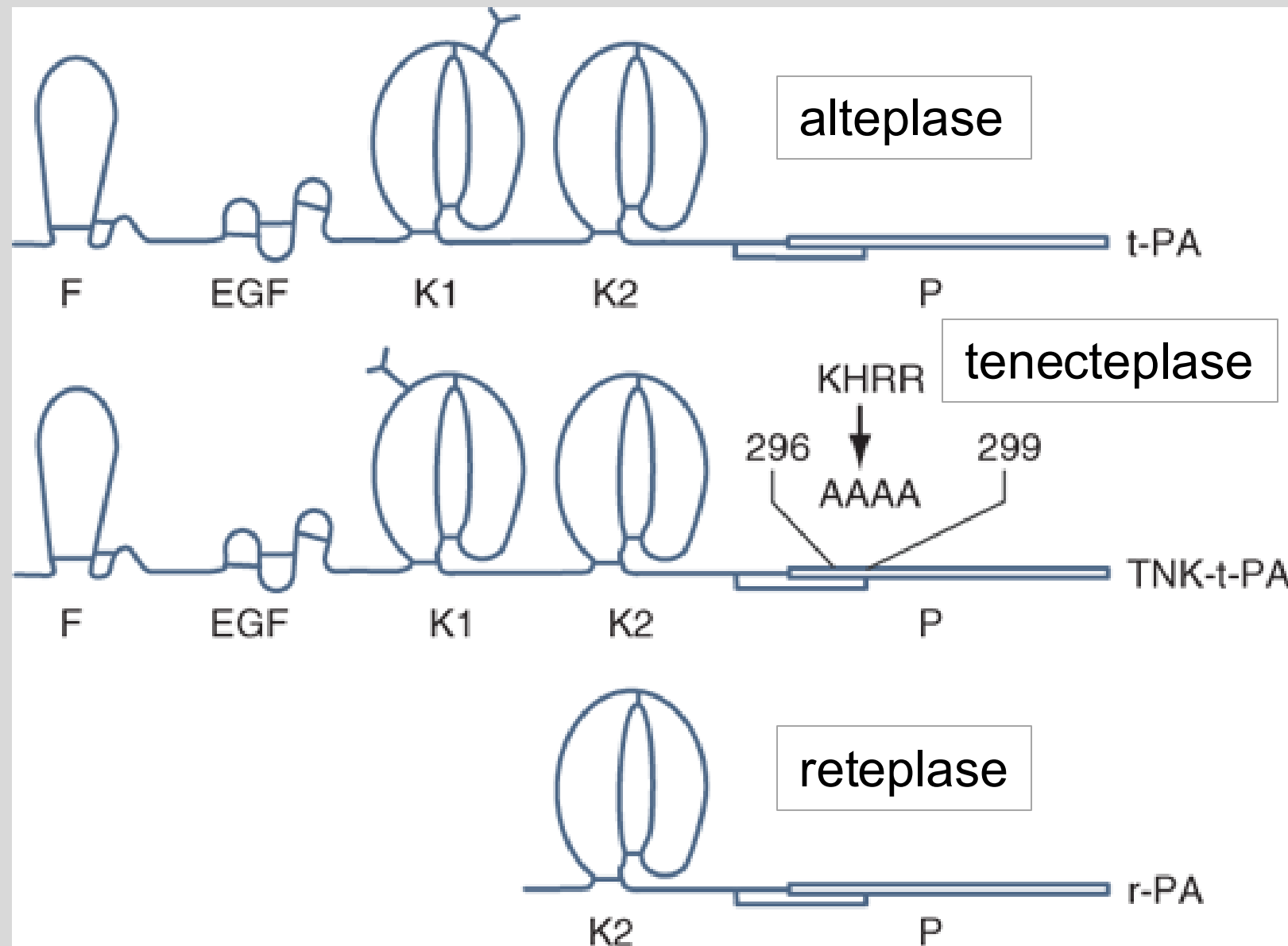
Key concepts: Antiplatelet agents

- Platelet aggregation inhibitors decrease the formation of a platelet-rich clot or decrease the action of chemical signals that promote platelet aggregation.
- The antiplatelet agents in this lecture block platelet COX-1 synthesis of thromboxane (aspirin), inhibit purinergic P2Y₁₂ ADP receptor (clopidogrel, prasugrel, ticagrelor, cangrelor), and block GP IIb/IIIa receptors (eptifibatide, tirofiban).
- Low-dose aspirin is used for primary and secondary prevention myocardial infarction transient cerebral ischemia. It may be used with another antiplatelet agent
- P2Y₁₂ receptor blockers are used for prevention of thrombotic events in patients with recent MI or stroke, percutaneous coronary intervention (PCI), acute coronary syndrome, and established peripheral artery disease.
- GP IIb/IIIa inhibitors are intravenous adjuncts to PCI.
- As with all drugs that interfere with platelet function or coagulation, bleeding is the most important adverse effect of this class of drugs.

Key concepts: Reversal Agents

- Protamine sulfate is a highly cationic protein that combines and forms a stable complex with heparin and neutralizes it.
- Vitamin K (phytonadione) reverses warfarin anticoagulation by outcompeting warfarin at VKORC1. It is administered orally or IV for severe bleeding.
- Andexanet-alfa is inactivated factor Xa for reversal of anticoagulation by the oral direct factor Xa inhibitors (DOACs) rivaroxaban and apixaban.
- Idarucizumab binds specifically to dabigatran and its active acylglucuronide metabolites, reversing the anticoagulant activity.
- 4-factor PCC from human plasma consists of unactivated factors II, VII, XI and X, Protein C, and Protein S. It is administered for urgent reversal of bleeding due to acquired coagulation factor deficiency induced by vitamin K antagonists and DOACs.

Structure-activity relationships of t-PA fibrinolytics



Source: Joseph Loscalzo, Anthony Fauci, Dennis Kasper, Stephen Hauser, Dan Longo, J. Larry Jameson: Harrison's Principles of Internal Medicine, 21e
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- Alteplase (t-PA)
Recombinant version of natural protein
- Tenecteplase:
 1. K1 glycosylation site (Y) repositioned → ↑ $t_{1/2}$
 2. Protease domain substitution → resistance to PAI-1 inhibition
- Reteplase:
Nonglycosylated, truncated variant
Longer $t_{1/2}$ than alteplase
Binds fibrin more weakly



Harrison's, 21e; Figure 118-9. F: finger; EGF: epidermal growth factor
K1, K2: first and second kringle domains, respectively); P: protease domain

Pharmacology of Fibrinolytic Agents: tPA and Derivatives

PK	IV, rapid hepatic clearance Alteplase $t_{1/2} < 5$ min Tenecteplase $t_{1/2} \sim 20$ min, $t_{1/2\text{terminal}} 90-130$ min Reteplase $t_{1/2} \sim 15$ min
MOA	Fibrinolytic agents catalyze the formation of plasmin from its precursor, plasminogen. The drugs induce rapid dissolution of thrombi. Both protective hemostatic thrombi and target thromboemboli are broken down, which can cause a generalized lytic state → hemorrhage
Uses	1- Acute ischemic stroke to ↓ neuronal death / infarction admin within 4.5 h 2- STEMI if PCI cannot be performed within 120 min, admin within 30 min of first medical contact; may still consider in patients who present late and have ongoing ischemia, admin within 12-24 h of symptom onset (with antiplatelet and anticoagulant) 3- PE treatment 4- Central venous catheter / dialysis catheter clearance (alteplase)
AE	Hemorrhage / hemorrhagic stroke, the major toxicity, resulting from lysis of hemostatic plugs at site of injury and from systemic plasmin generation, which degrades coagulation factors (esp. V and VII) as well as fibrin.

Absolute and Relative Contraindications to Fibrinolytic Therapy

ABSOLUTE CONTRAINDICATIONS	RELATIVE CONTRAINDICATIONS
Prior intracranial hemorrhage	Uncontrolled hypertension (SBP >180 mm Hg or DBP >110 mm Hg)
Known structural cerebral vascular lesion	Traumatic or prolonged CPR or major surgery within 3 weeks
Known malignant intracranial neoplasm	Recent (within 2-4 weeks) internal bleeding
Ischemic stroke within 3 months	Noncompressible vascular punctures
Suspected aortic dissection	For streptokinase: prior exposure (>5 days ago) or prior allergic reaction to streptokinase
Active bleeding or bleeding diathesis (excluding menses)	Pregnancy
Significant closed-head trauma or facial trauma within 3 months	Active peptic ulcer
Goodman & Gilman 13e Table 32-3 CPR, cardiopulmonary resuscitation INR, international normalized ratio	Current use of warfarin and INR >1.7

Aspirin Pharmacokinetics and Mechanism of Antiplatelet Action

Aspirin is taken up by platelets in the portal circulation before it is deacetylated to salicylate on the first pass through the liver.

- Oral
- ASA $\xrightarrow{\text{esterases}}$ salicylate
 - gut mucosa
 - blood
 - red blood cells
 - synovial tissues
- Salicylate: hepatic conjugation
- Renal excretion of
 - salicyluric acid (75%)
 - salicylic acid (10%)
- $t_{1/2}$ in portal circulation ~20 min

Mechanism:

Aspirin irreversibly acetylates an allosteric site

↓
Inhibits TXA₂ synthesis

↓
TXA₂ actions are prevented

↓
reduces platelet aggregation and
vasoconstriction

- Effects lasts for the life of the platelet (7-10 days).
- **Thus, repeated LOW doses of aspirin produce a cumulative effect on platelet function.**

Aspirin's Antiplatelet Uses and Cautions

Dual therapy with P2Y12 receptor inhibitor for some indications.

Therapeutic Uses:

Cardiovascular protection to reduce the risk of thrombotic events – heart attack and ischemic stroke – in high-risk patients

Secondary prevention:

- To reduce risk of subsequent CV disease events

Primary prevention

- High-risk patients with established CVD
- Intermediate- and high-risk in discussion with patient regarding individual preferences and risk/benefit

Adjunct in revascularization procedures, such as PCI, CABG, stenting

PCI: percutaneous coronary intervention;
CABG: Coronary artery bypass graft

Cautions:

- Upper gastrointestinal bleeding
- Surgery: Avoid use 1-2 weeks prior to surgery except if patient has previous cardiac stent placement
- Hypersensitivity: Aspirin-induced urticaria, rhinosinusitis, bronchospasm, anaphylaxis
- Pregnancy: Low doses for certain medical conditions have not been shown to cause fetal harm.

Recommended to discontinue prior to delivery.

- Children with viral illness: Association with Reye's syndrome

Irreversible Inhibitors			Reversible Inhibitors	
A D M E t_{1/2}	Clopidogrel	Prasugrel	Ticagrelor	Cangrelor*
	Oral prodrug Onset Day 2	Oral prodrug Onset <30 min	Oral Onset ~2 h	IV bolus f/b infusion T _{max} <2 min
	Protein binding 98%	98% (active metab)	>99%	98%
	CYP2C19 conversion to active thiol metab (15% of dose)	Hydrolysis then CYP3A4, 2B6 to active thiol metab	CYP3A4 active thiol metab	Rapid dephosphorylation in plasma, inactive
	Urine, feces	Urine, feces	Urine, feces	Urine, feces
	Clopidogrel ~6 h Active metab ~30 min Platelet bound ~11 dys	Active metab ~7 h	Ticagrelor ~24 h after drug is discontinued	3-6 min Rapid return of platelet function

Activation of prasugrel is more efficient →
faster onset of action and greater, more
predictable inhibition of ADP-induced platelet
aggregation

*Cangrelor is an ADP
analog

Therapeutic Uses / Adverse Effects

Irreversible Inhibitors		Reversible Inhibitors	
Clopidogrel and Prasugrel		Ticagelor	Cangrelor
Dual therapy with aspirin Secondary CV protection to reduce risk of thrombotic events • STEMI: following MI with or without reperfusion therapy • Acute coronary syndrome: After non-STEMI or in unstable angina • PCI: After bare metal stent or drug-eluting stent placement • Patients intolerant to aspirin after STEMI		• Acute coronary syndrome: patients with a history of MI or unstable angina • Coronary artery disease, stable (primary prevention) • Minor ischemic stroke	• Adjunct to PCI to patients who have not been treated with a P2Y₁₂ inhibitor or GP IIb/IIIa inhibitor • Bridging therapy prior to cardiac surgery following thienopyridine discontinuation
Adverse effects: All antiplatelet drugs are associated with bleeding.			
CYP2C19 poor metabolizers → ↓ conversion to active thiol	Bleeding, higher rates than with clopidogrel • Not recommended in patients ≥75 years	• Thrombocytopenic purpura • Bradyarrhythmia • Dyspnea (resolves)	Bleeding

GP IIb/IIIa Inhibitors: Pharmacology

Eptifibatide	Tirofiban
Cyclic heptapeptide	Synthetic nonpeptide
t _{1/2} ~2.5 hours	t _{1/2} ~2 hours
IV bolus	IV bolus
Platelet function restored in ~4 hours	Platelet function restored in 4-8 hours

GP IIb/IIIa Inhibitors reversibly block platelet aggregation induced by any agonist.

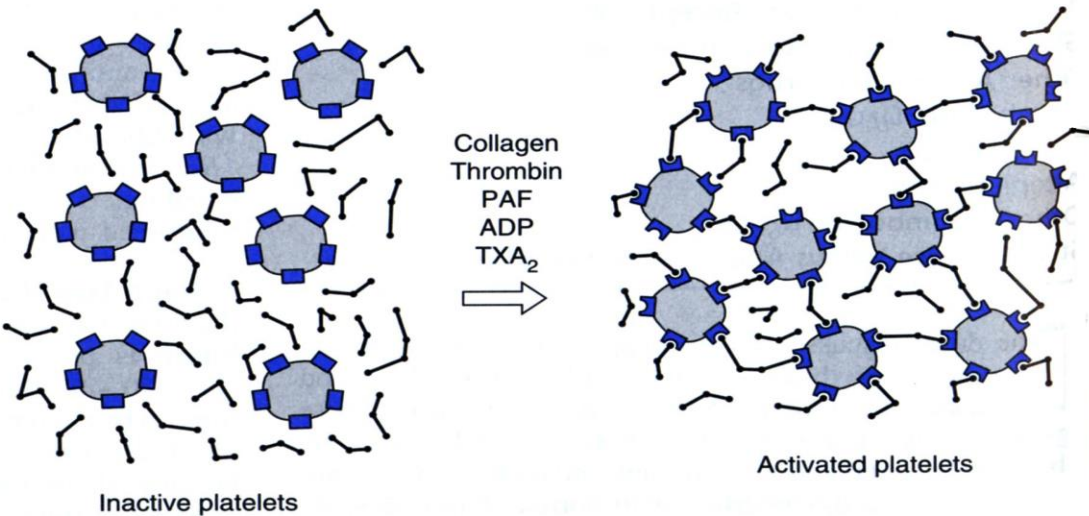
↓
reversibly block fibrinogen binding and bridging of platelets

↓
prevent platelet aggregation

Contraindications

- | | |
|--|---|
| <ul style="list-style-type: none">• Active abnormal bleeding within the previous 30 days• History of bleeding diathesis• History of stroke within 30 days• History of hemorrhagic stroke• PT >1.2 times control or INR ≥2.0 | <ul style="list-style-type: none">• Severe hypertension not adequately controlled on medication• Major surgery within the preceding 6 weeks• Dependency on hemodialysis• History of intracranial disease• Thrombocytopenia• Hypersensitivity |
|--|---|

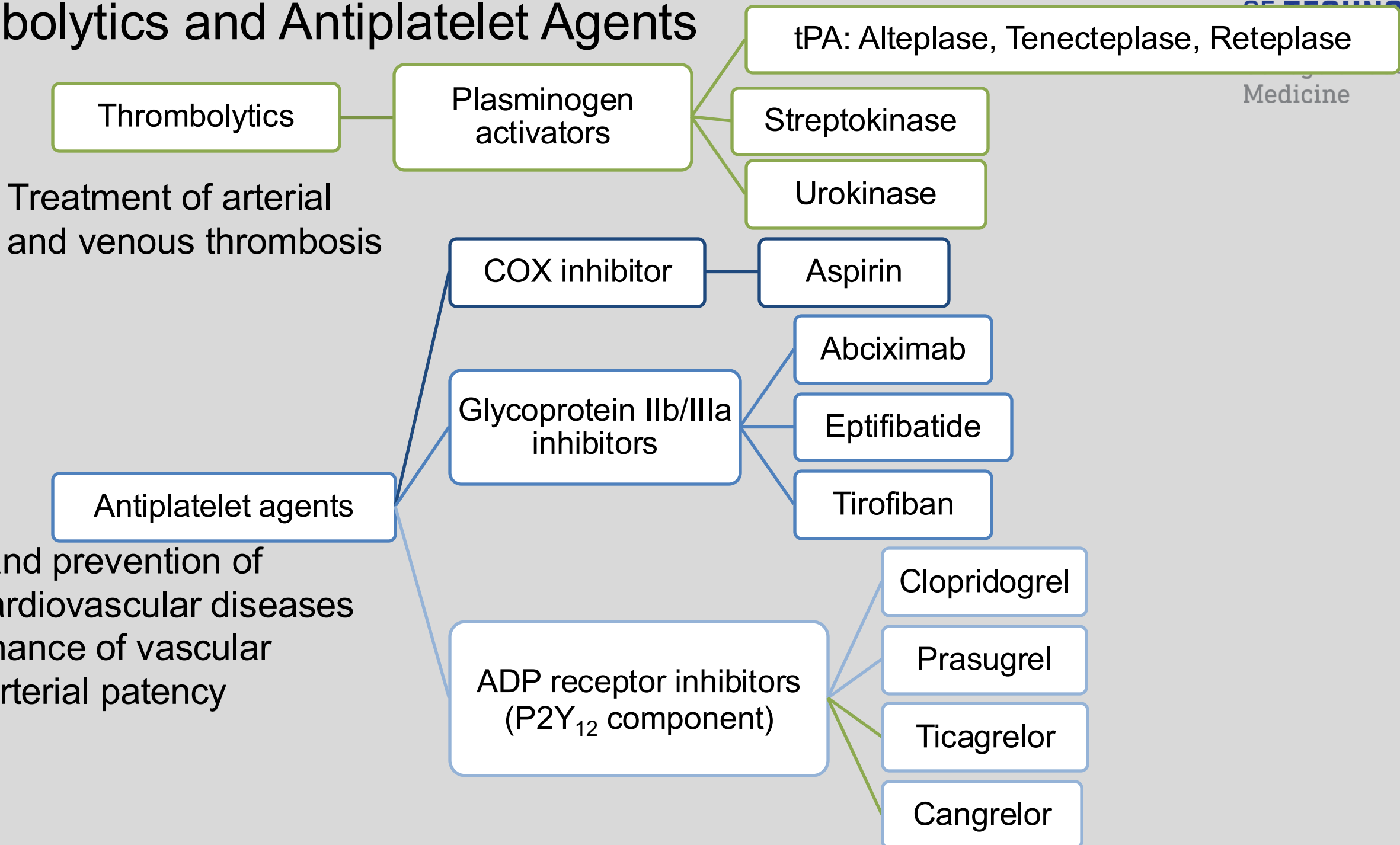
B Platelet Aggregation



Agents for Reversal of Anticoagulant Activity

Antidote	Anticoagulant	Mechanism
Protamine	Heparin	Highly cationic protein combines with acidic UHF and LMWH forming stable complex that is excreted • Acts in ~5 minutes; $t_{1/2}$ ~7 minutes • Protamine has weak anticoagulant effects, therefore excessive protamine dosing may worsen bleeding.
Vitamin K Phytonadione	Warfarin	Outcompetes warfarin at VKORC1 • Oral onset of action 6-10 hours; INR returns to normal in 24-48h; may cause warfarin resistance
Prothrombin complex concentrate (also FFP)	Urgent reversal of VKAs and DOACs	4-factor PCC increases the levels of vitamin K-depended coagulation factors II, VII, IX, and X plus protein C and S • May predispose the patient to a thromboembolic complication (weigh risk v benefit)
Idarucizumab	Oral direct thrombin inhibitor	MAb fragment that specifically binds dabigatran and its active acyglucuronide metabolites • Onset in minutes; duration 24h or more

Thrombolytics and Antiplatelet Agents



SUMMARY OF FIBRINOLYTIC AGENTS

Streptokinase (n/a in U.S.)	• 47,000 Da protein produced by β-hemolytic streptococci • It has no intrinsic enzymatic activity • 1:1 complex with plasminogen $\rightarrow$ conformational change $\rightarrow$ exposes plasminogen's active site $\rightarrow$ complex cleaves circulating (free) plasminogen $\rightarrow$ forms plasmin • Activates both free and fibrin-bound plasminogen $\Rightarrow$ lytic state • Cross reactivity with circulating anti-streptococcal antibodies $\rightarrow$ may cause allergic reactions or neutralize the activity of streptokinase
Urokinase	• Human enzyme synthesized by the kidney • Directly converts plasminogen to plasmin • Activates both free and fibrin-bound plasminogen
Alteplase ($t_{1/2}$ 5 min)	• Recombinant t-PA • Preferentially activates plasminogen bound to fibrin
Reteplase ($t_{1/2}$ 13-16 min)	• Recombinant t-PA variant; lacks the finger, EGF, and first kringle domains $\rightarrow$ more weakly binding of fibrin; longer $t_{1/2}$ (2 bolus doses)
Tenecteplase ($t_{1/2}$ 90-130 min)	• Recombinant t-PA variant; longer $t_{1/2}$ (1 bolus dose) • Relatively resistant to plasminogen activator inhibitor-1 compared to alteplase and reteplase

Summary of Antiplatelet Agents

- Aspirin acetylates cyclooxygenase-1 (COX-1) and inhibiting TXA2 synthesis. Low-dose aspirin effectively inhibiting platelet aggregation for 5 to 7 days (the life of the platelet).
- ADP receptor P2Y₁₂ inhibitors prevent activation of the GP IIb/IIIa receptor complex, thus preventing fibrinogen cross-linking. Clopidogrel and prasugrel (thienopyridines) are irreversible inhibitors. Reversible inhibitors are ticagrelor (oral) and cangrelor (I.V.)
- GP IIb/IIIa inhibitors abciximab, eptifibatide, and tirofiban block platelet aggregation by blocking the binding of fibrinogen, thereby preventing fibrinogen cross-linking.